Rethinking BDM Belief Elicitation and Incentive Compatibility

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This Project

I experimentally study whether information on incentives in the BDM belief elicitation mechanism increases truthful reporting by subjects

Belief Elicitation

Belief Elicitation is an integral part of an economist's toolkit

- Deviations in games (Bellemare et al., 2008; Costa-Gomes and Weizsacker, 2008)
- Trust and prosocial behavior (Ashraf et al., 2006; Costa-Gomes, et al., 2015; Zimmermann and Schwerter, 2022)
- Macroeconomics (Guiso and Parigi, 1999; Branch, 2004; Engelberg et al., 2011)
- Belief formation and updating (Coffman et al., 2019; Zimmermann, 2020; Barron, 2020)

Popular Mechanisms

Popular belief elicitation mechanisms

- Quadratic Scoring Rule (QSR. Brier, 1950)
- Binarized Scoring Rule (BSR. Hossain and Okui, 2013)
- Becker-DeGroot-Marschack (BDM. Becker et a., 1964; DuCharme and Donnell, 1973; Karni, 2009)
  - BDM belief elicitation is particularly popular

How to Choose?

All of the above mechanisms satisfy the property that we call *Theoretical Incentive Compatibility* (TIC)

Definition (Theoretical Incentive Compatibility)

A belief elicitation mechanism is TIC if given the incentives, the dominant strategy for subjects is to report their beliefs truthfully.

Each mechanism needs some additional assumptions to satisfy TIC

- BSR and QSR need *EU*
- QSR needs *risk neutrality*
- BDM *does not need either*

Motivation

The popularity of BDM: a non-exhaustive list of papers that use BDM to incentivize belief elicitation

- Drobner (2022): motivated beliefs
- Kim and Kim (2022): game theory
- Castagnetti and Schmacker (2022): motivated beliefs
- Dustan et al. (2022): gender
- Casoria et al. (2022): social identity & discrimination
- Yaouanq and Schwardmann (2022): present bias
- Guo and Recalder (2022): gender
- Mobius et al. (2021): motivated beliefs
- Kessel et al. (2021): gender
- Valero (2021): motivated beliefs
- Barron (2020): motivated beliefs
- Aguirregabiria and Xie (2020): game theory
- Coutts (2019): belief updating
- Buser et al. (2018): belief updating

Motivation

Is TIC as good as it sounds?

- Recent evidence shows that TIC mechanisms may not perform as advertised in practice
  - Danz et al. (2022) shows that BSR incentives actually leads to more false reports
- We need to think about a different type of IC: Behavioral Incentive Compatibility (BIC)

Behavioral Incentive Compatibility

Definition (Behavioral Incentive Compatibility (Danz et al., 2022))

A BIC mechanism satisfies the following two conditions:

1. Information on incentives increases truthful reporting
2. When given a choice over pure incentives, subjects select the outcome that's uniquely maximizing under the mechanism

What does it mean?

- Misreporting from confusion over the mechanism does not necessarily mean BIC is violated (Cason and Plott, 2014)
- Condition 2 may not apply generally. In Danz et al. (2022), BSR can be reduced to a choice over two compound lotteries

Questions

- Does the BDM belief elicitation mechanism induce truthful reporting?
- If not, what are the factors that result in misreporting?
  - Subject confusion?
  - Violation of BIC?

What is BDM?

We are interested in the subject's belief of event E, $\pi(E)$

- The subject is asked to report their belief $\mu \in [0, 1]$
- A random number r is generated from the uniform distribution $[0, 1]$
  - If $r \leq \mu$, the mechanism awards the subject the event bet $x_E y$ where $x > y$ (event lottery)
  - If $r > \mu$, the subject is awarded the number bet $l(r, x, y)$ (number lottery)

Overview of BDM

TIC of BDM requires 3 assumptions

Assumption (Probabilistic Sophistication)

A subject's preference relation $\succsim$ over the set of finite acts and lotteries D is said to exhibit probabilistic sophistication if it ranks acts or lotteries solely on the basis of their probability distributions over outcomes. *(Machina and Schmeidler, 1995)*

This means that for all acts f and lotteries $l(p, x, y) = [x, p; y, (1-p)]$, $p \in [0, 1]$, $\pi(f^{-1}(x)) = p$ implies $x_{f^{-1}(x)} y \sim l(p, x, y)$

Overview of BDM

What does it mean for an agent to be probabilistically sophisticated?

- The agent **can** evaluate subjective probabilities (and does so)
- Preferences **need not** satisfy expected utility
- The agent is **not** ambiguity averse
- The agent **cannot** have preferences for certain sources of ambiguity

Overview of BDM

Assumption (Dominance)

A subject's preference relation over the set of finite acts and lotteries D exhibits dominance if for $x > y$, $p \geq p'$ implies $l(p, x, y) \succsim l(p', x, y)$, with strict preference if $p > p'$.

- This is a much simplified version of Axiom 5 (First-Order Stochastic Dominance Preference) in Machina and Schmeidler (1995).
- Subjects ranks acts and lotteries only based on their expected payoffs and never prefers a money lottery that's FOSD'd

Overview of BDM

Assumption (No Stakes)

The No Stakes assumption requires that subjects have no financial stakes in the realization of event E apart from the incentive payments on the belief elicitation.

TIC of BDM

Why is BDM attractive?

- It does not require risk neutrality
- Strictly speaking, EU is not required (Dominance is weaker than Independence Axiom)
  - However, in our simple environment, probabilistic sophistication + dominance + reduction of compound lotteries = EU

Are there downsides?

- Difficult to explain
- Implementation can be cumbersome

Implementation of BDM

There are 4 main ways of implementing BDM:

- Direct elicitation (what I do)
- Random Binary Choice/MPL (Healy, 2020)
- Synchronized 2-Stage Lottery Choice Menu (Holt and Smith, 2016)
- Clock Auction (Karni, 2009; Hao and Hauser, 2012)

Testing BIC

Going back to the 2 weak conditions for BIC:

1. Information on deployed incentives increases truthful revelations
   - Generally testable, my experiment focuses on this
2. When given a choice over pure incentives, most participants select the outcome that is uniquely maximizing under the mechanism
   - Not universally testable. BSR can be reduced to 2 compound lotteries once belief is reported, but BDM cannot due to the randomization device

Design

Key Problem: We cannot read minds

How do we get around this?

- Elicit beliefs on events with objective priors
  - Subjects endowed with beliefs = objective priors
  - Compare reports against objective priors

Design

Main treatments

- Full Information
  - Subjects learn the full incentive scheme of BDM
- No Information
  - Subjects do not receive incentive information
  - Simply told that truth-telling is in their best interest
  - Bonus is still determined by BDM

Design

Prior (outcome of interest):

- 9 events (scenarios) with objective priors 10% to 90%
- Each scenarios has 10 urns that are either red or blue
- An urn is randomly selected at the beginning of each scenario and subjects guess the probability it's red
- Subjects know the number of red and blue urns

Design

Update:

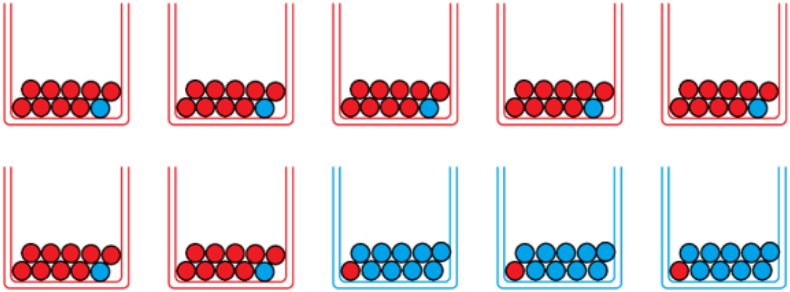
- Each urn contains 10 balls of red and blue colors
- Subjects know contents of urns
- All red urns have the same contents, so do all blue urns
- Red urns have majority red balls, similarly for blue urns
- After subjects report the prior, a ball is randomly drawn and subjects update their guess

Purpose:

- Less suspicious
- Do incentives matter for tasks that require more cognitive effort?

Design

Example Scenario:



- In each question, the diagram for the scenario is shown
- Full instructions (in both treatments) and incentives (in Full Info treatment) also available

Design

Elicitation Interface:

What is the chance that the selected urn is red?

Please use the slider below to indicate what you believe are the chances out of 100 that the selected urn is **red**.

Click the button below if you need to review the information for this scenario again.

Show/hide information

0 10 20 30 40 50 60 70 80 90 100

Chances out of 100 that the selected urn is red

Design

Incentives:

- Both treatments are incentivized, each question can earn a bonus of $0.5
- Bonus is calculated using BDM in both treatments
- Subjects are told truthful reporting maximizes chance of winning bonus
- Only difference between treatments is whether quantitative incentives are shown

Design

Explanation of BDM Incentives: Text

Here is how the payment rule works:

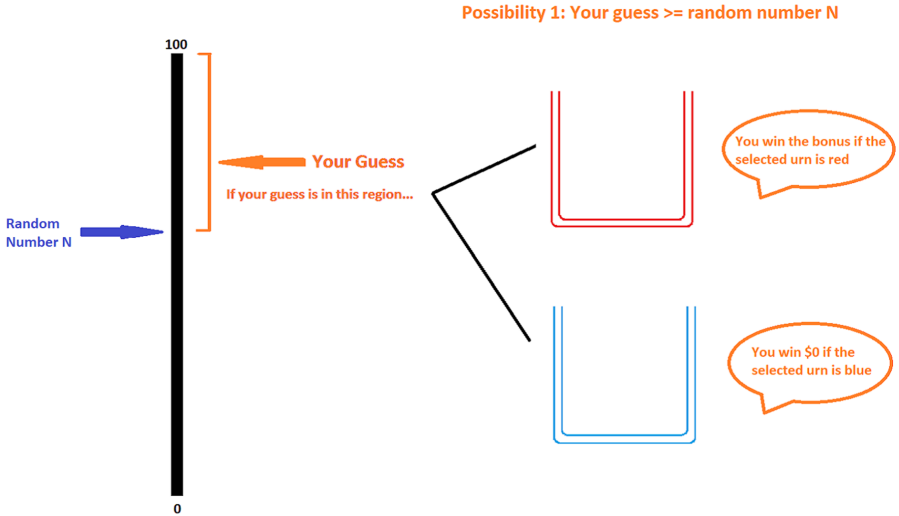
After you report your guess in a question, the computer will use your guess to enter you into one of two lotteries: the **Event Lottery** and the **Number Lottery**. Your bonus earning for this question will be determined by playing the lottery you are entered into. The details are explained below.

First, the computer randomly draws a number N between 1 and 100. All whole numbers between 1 and 100 have equal chances of being drawn. After you report your guess, the computer compares this random number N and your reported guess, and

- **If your guess is greater than or equal to the random number N**, the computer will enter you into the **Event Lottery**. When you play this lottery, you will earn the bonus of $ for this question if the selected urn in this question is indeed red. If the selected urn in this question is blue, you will earn $0 for this question. (For example, this means that *if you report 100, you will always be paid by the Event Lottery*.)
- **If your guess is less than the random number N**, the computer will enter you into the **Number Lottery**. When you play the Number Lottery, your bonus earnings for this question depends on random chance. This lottery gives you N chances out of 100 to earn the bonus of $, and with (100-N) chances out of 100, you will earn $0. (For example, if the random number N is 70 and your reported guess is 50, then you will play the lottery which gives you a 70% chance to win the bonus, and a 30% chance to win $0. And *if you report 0, you will always be paid by the Number Lottery*)

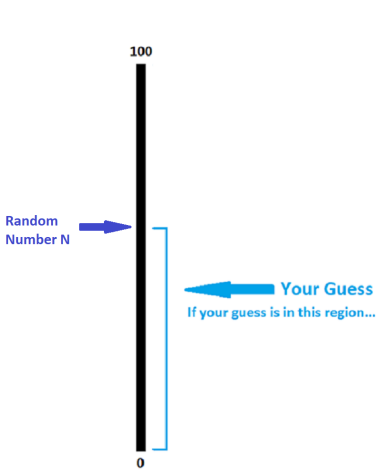
Design

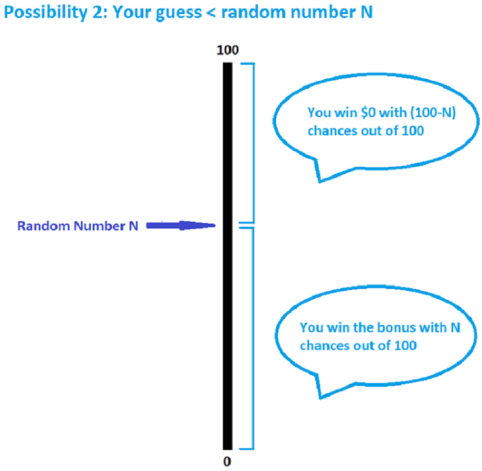
Explanation of BDM Incentives: Diagram for Event Lottery



Design

Explanation of BDM Incentives: Diagram for Number Lottery





Data

- Study conducted on Prolific
- First round of data collection in October 2022
- 99 subjects over 2 treatments

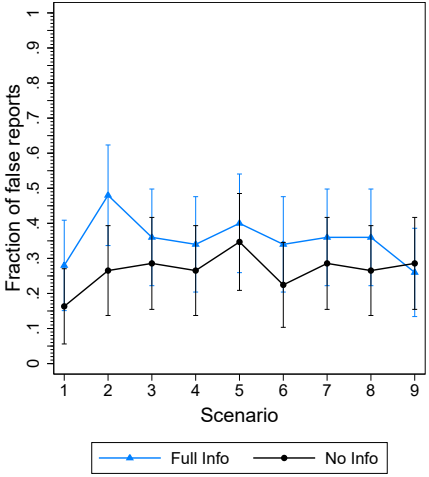
Data

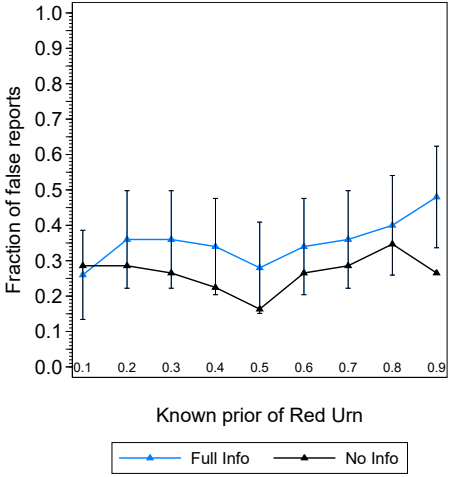
Sample Summary Statistics

	Full Info		No Info		Overall	
	mean	sd	mean	sd	mean	sd
Age from Prolific	36.73	11.62	34.69	11.00	35.71	11.31
Gender: Woman	0.46	0.50	0.65	0.48	0.56	0.50
Gender: Man	0.52	0.50	0.31	0.47	0.41	0.50
Bachelor's degree or above	0.44	0.50	0.47	0.50	0.45	0.50
Taken some probability/statistics course	0.44	0.50	0.39	0.49	0.41	0.50
Total approvals on Prolific	1278.78	962.20	1107.47	836.11	1193.12	900.83
Observations	50		49		99	

Results: Priors

Result 1: Percentage of false reports in Full Info treatment is around 9 percentage points higher than the No Info treatment, however, the difference is not significant





Results: Priors

Pooled OLS regression over all scenarios, SE clustered at individual level

	False Report for Prior
full_info	0.0880
	(0.0737)
_cons	0.265***
	(0.0492)
N	891

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Probit model gives the same results.

Results: Priors

Is the higher percentage of false reports in the Full Info treatment driven by confusion?

- Classify subjects by whether they understood the incentives
- Use the 4 questions that check incentive understanding at the end of survey

Results: Priors

Sample understanding check question 1:

Suppose in hypothetical question of a scenario, the randomly drawn number is 20, and you reported a guess of 15. Please answer the following using this information. You can see the payment rule here for more reference.

How will your bonus payment be determined?

Event Lottery

Number Lottery

Results: Priors

Sample understanding check question 2:

What are your chances of winning the bonus for this question?

- I win the bonus with 20% chance
- I win the bonus with 80% chance
- I win the bonus with 85% chance
- I win the bonus with 15% chance
- I win the bonus if the selected urn is red
- I win the bonus if the selected urn is blue

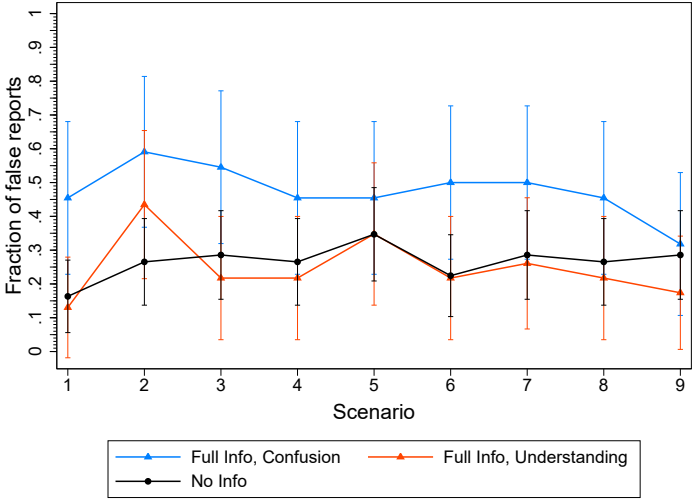
Results: Priors

Classification of understanding:

- If passed all 4 incentive understanding check questions, classify as understanding incentives
- If answered 1 or more incorrectly, classify as did not understand incentives

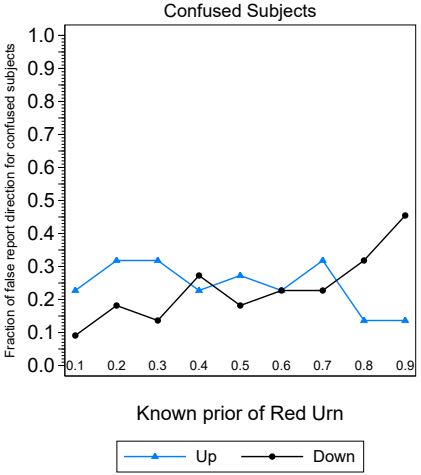
Results: Priors

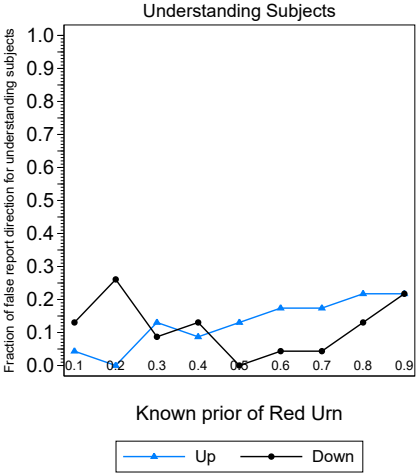
Result 2: False reports in priors in the Full Info treatment are driven by subjects who did not understand incentives.



Results: Priors

Result 3: False reports by subjects who did not understand BDM incentives pull to the center.

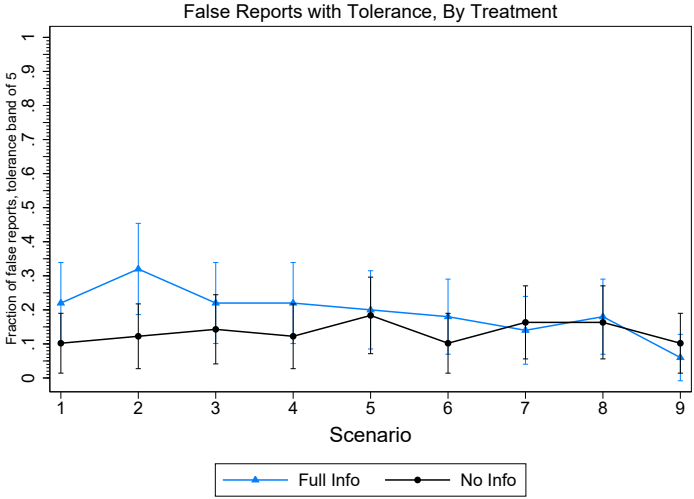




Results: Priors

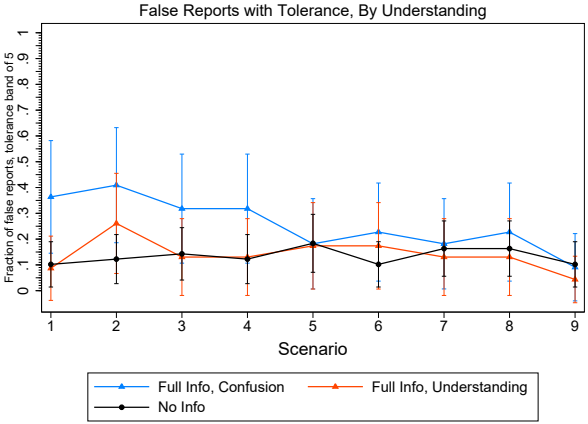
Q: Are the false reports noise from the slider instrument?

A: Some, but not all.



Results: Priors

Result 2 revisited

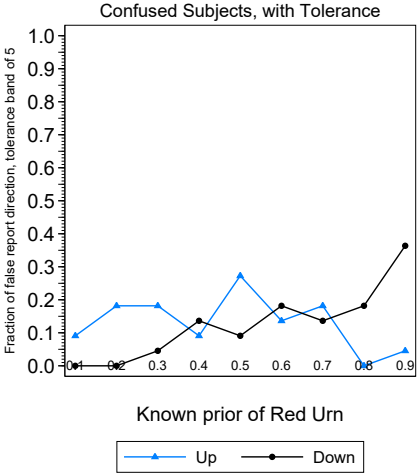


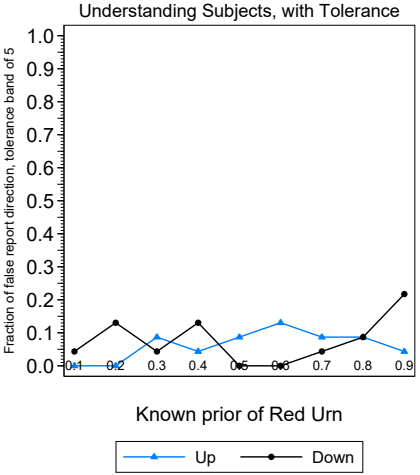
Conjectures:

- Learning
- Attention checks affecting subject behavior

Results: Priors

Result 3 revisited





Results: Priors

Magnitude of false reports

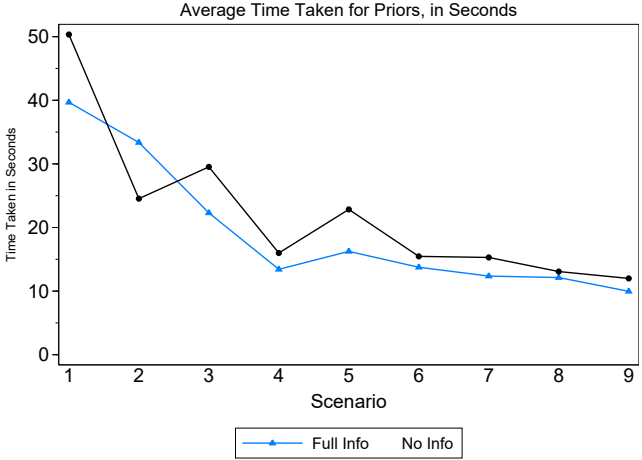
Table: Mean Absolute Deviation in False Reports

	Strict Definition		With Tolerance Band of 5	
	mean	sd	mean	sd
full_info	3.72	7.68	16.82	9.22
no_info	2.66	7.23	17.44	11.50
Total	3.19	7.48	17.07	10.17

Results: Priors

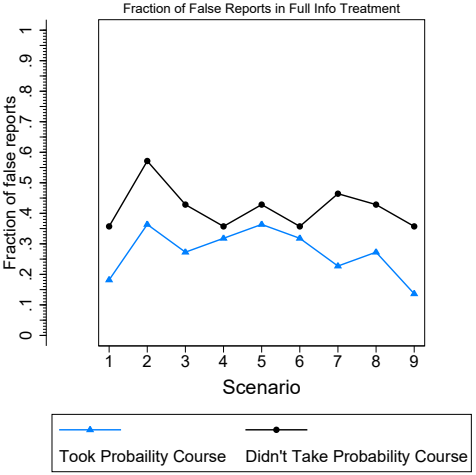
Most belief elicitations are more complicated than scenarios with endowed priors

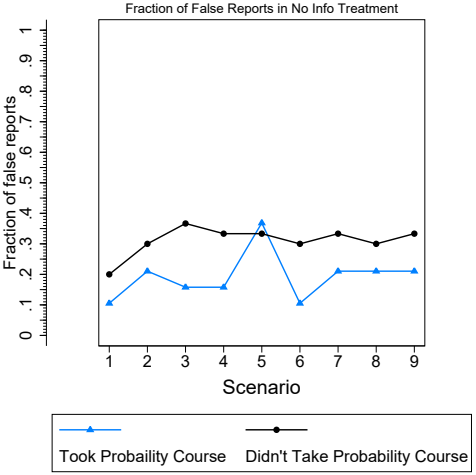
Result 4: Revealing incentives did not induce more effort in prior reports



Results: Priors

Do subjects with a better understanding of probability perform better?





Results: Updates

Result 5: No one is fully Bayesian.

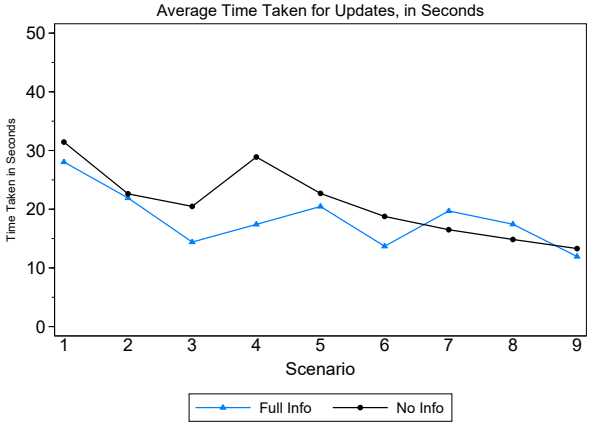
Treatment	False Report for Posterior
Full Info	50
No Info	49
Total	99

Results: Updates

Q: Did incentives induce more effort on the more complicated task of belief updating?

A: No.

Result 6: Revealing full incentives did not induce more effort on the updating task.

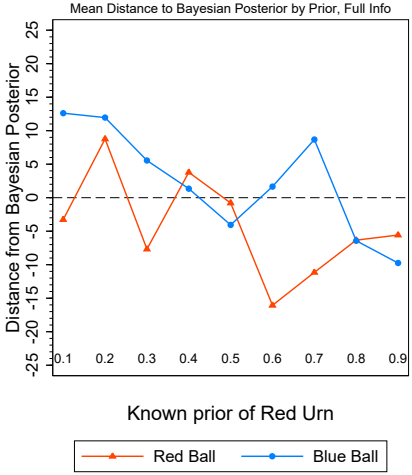


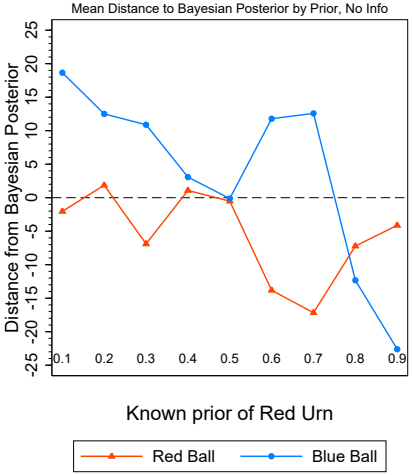
Results: Updates

- Incentives did not affect the quality of updates (deviation from Bayes, correct direction)
- Probability coursework did not affect the quality of updates
- Understanding incentives did not affect updates
- None of these factors influenced effort

Results: Updates

Most subjects are conservative in their updates





Conclusion

Bottom line: BDM is more complicated to implement, harder to understand, and does not perform better

- Instructing subjects on incentives is more work for both the experimenter and the subject
  - In the Full Info treatment, mean time spent on the incentives page ~ 2.5 minutes (sd 3 min, max 20 min!)
  - In the No Info treatment, mean time spent is 25 seconds (max 1.3 minutes)

Conclusion

Subject feedback

'The bonus payout was a bit complicated - I did try to determine if there was a way to maximize my bonus - but just decided to give each question my best guess'

'Totally convoluted directions and way to much to digest and memorize for the pay. Honest feedback, Id skip the lottery part unless there is some purpose to it Im unaware of.'

Conclusion

Some lessons for experimental design

- Do not use sliders for belief elicitation
- Be careful about attention check placement

Appendix

What is QSR

- Continuous payoff function
- Quadratic loss function
- Payoff is $[a - b(1-\mu)^2]$ if event obtains $(a > b > 0)$
- Payoff is $(a - b\mu^2)$ if event does not obtain

What is BSR

- Binary payoff
- Same loss function as QSR
- Receives payoff with probability $[1 - (1 - \mu)^2]$ if event obtains
- Receives payoff with probability $(1 - \mu^2)$ otherwise

Implementation of BDM

Random Binary Choice (Healy, 2020):

Q#		**Option A**		**Option B**
1	Would you rather have:	\$20 if E occurs	or	1% chance of \$20
2	Would you rather have:	\$20 if E occurs	or	2% chance of \$20
3	Would you rather have:	\$20 if E occurs	or	3% chance of \$20
⋮	⋮	⋮	⋮	⋮
99	Would you rather have:	\$20 if E occurs	or	99% chance of \$20
100	Would you rather have:	\$20 if E occurs	or	100% chance of \$20

- A row is randomly selected and the choice is implemented
- Switching point is the belief of the event
- Downside: almost impossible to implement

Implementation of BDM

Two-Stage Synchronized Lottery Choice Menu (SC) (Holt and Smith, 2016):

- Two-stage version of RBC/MPL
- First present a coarse table, present a finer table depending on the previous answer
- Less burdensome than the full MPL but difficult to explain

Implementation of BDM

First stage of SC:

Option A: Earn $4 if the top scorer in the group has odd ID

Option B: Earn $4 if the number on the ball drawn from Basket B is in the range indicated on each line.

	A	B	
Option A	○	○	1-10
Option A	○	○	1-20
Option A	○	○	1-30
Option A	○	○	1-40
Option A	○	○	1-50
Option A	○	○	1-60
Option A	○	○	1-70
Option A	○	○	1-80
Option A	○	○	1-90

Implementation of BDM

Second stage of SC (if subject switched at 20 in the first stage):

Option A: Earn $4 if the top scorer in the group has odd ID

Option B: Earn $4 if the number on the ball drawn from Basket B is in the range indicated on each line.

	A	B	
Option A	○	○	1-11
Option A	○	○	1-12
Option A	○	○	1-13
Option A	○	○	1-14
Option A	○	○	1-15
Option A	○	○	1-16
Option A	○	○	1-17
Option A	○	○	1-18
Option A	○	○	1-19

Implementation of BDM

Clock Auction (Karni, 2009; Hao and Hauser, 2012):

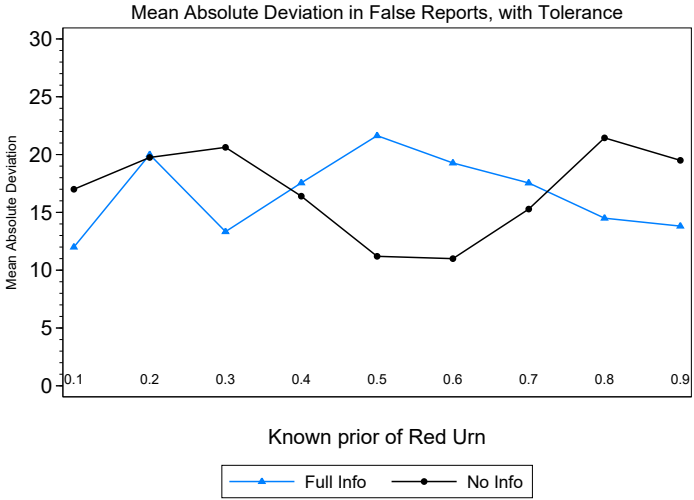
- Subject plays in an English auction against a robot bidder who exits at a randomly drawn number $r \in [0, 100]$
- If the subject exits first, they are rewarded the lottery $l(r, x, 0)$
- If the robot exits first, subject is rewarded $x_E 0$
- The subject's exit point is $\pi(E)$
- More intuitive to explain, but subject beliefs are censored if robot exits first

So what exactly is going on?

- Machina and Schmeidler (1992) introduced a formal characterization of probabilistically sophisticated **non-expected utility** preferences by modifying the Savage axioms
  - Remove P2-Sure Thing Principle
  - Modify P4-Weak Comparative Probability into P4*-Strong Comparative Probability
- Machina and Schmeidler (1995) present an alternative derivation of subjective probability without expected utility by introducing an objective randomizing device (roulette lotteries)
  - Axioms: Ordering, Mixture Continuity, FOSD Preference, Horse/Roulette Replacement

Results: Priors

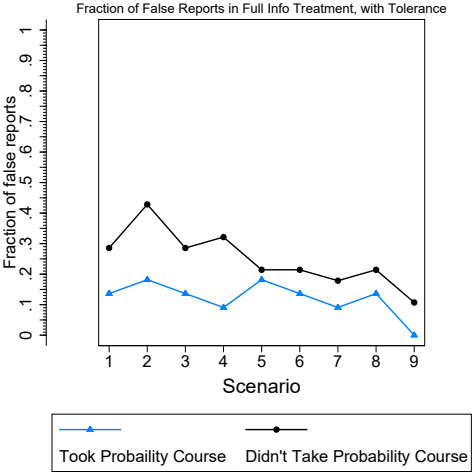
Mean absolute deviation in false reports from objective priors

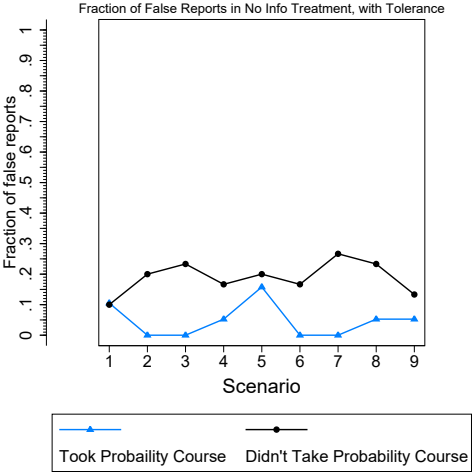


Results: Priors

Do subjects with a better understanding of probability perform better?

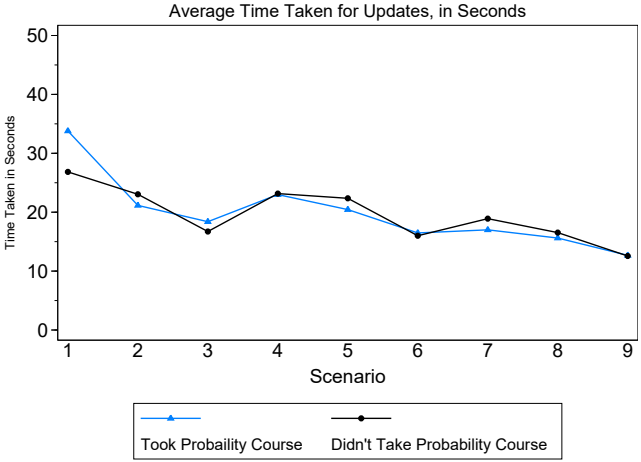
False reports defined with a tolerance band of 5





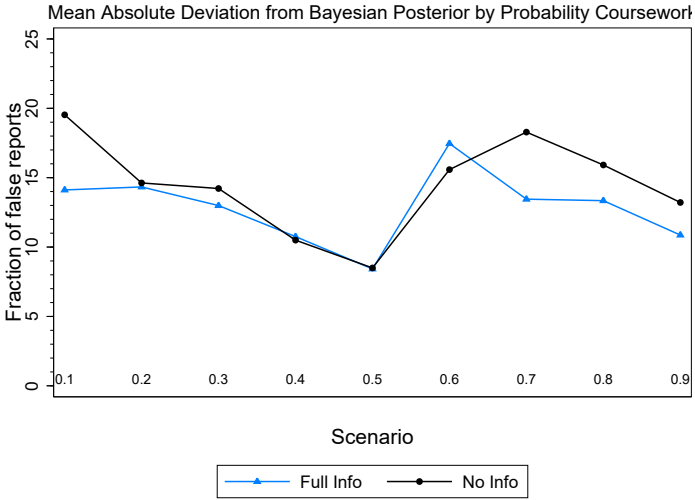
Results: Updates

Understanding of probability did not impact effort.



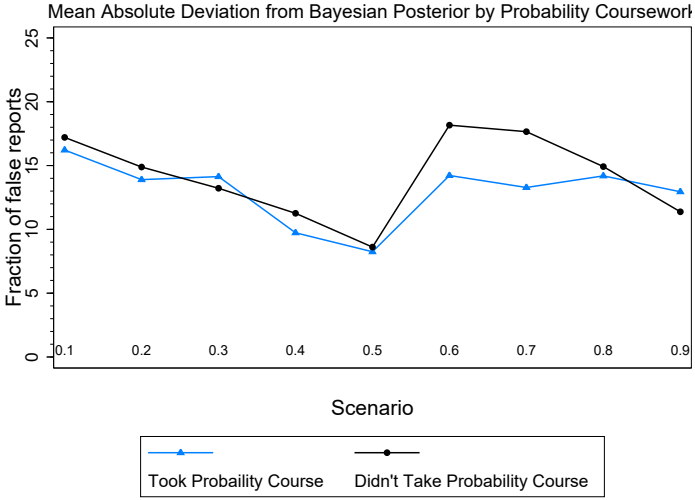
Results: Updates

Incentives did not affect mean absolute deviations from Bayesian posterior



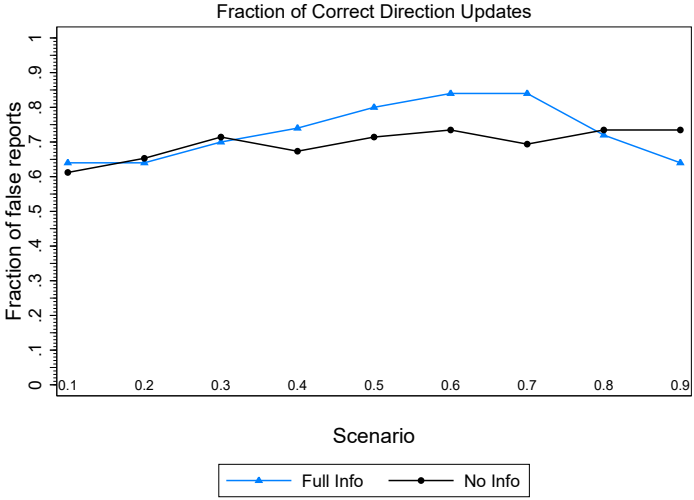
Results: Updates

Probability coursework did not affect mean absolute deviations from Bayesian posterior



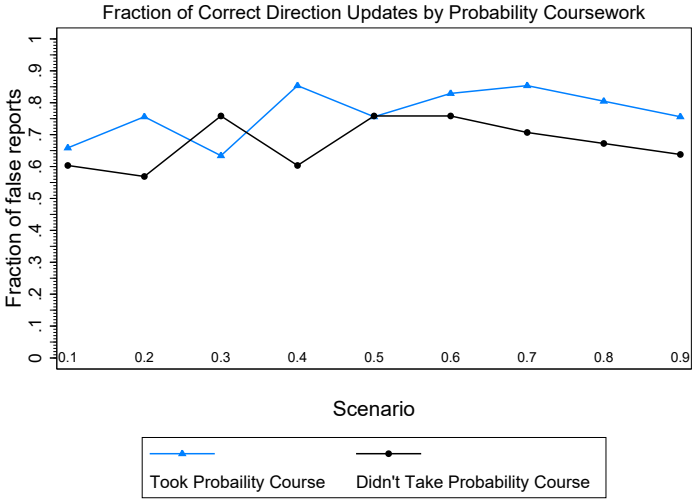
Results: Updates

Incentives did not impact whether subjects updated in the correct direction.



Results: Updates

Probability coursework did not affect direction of updates



Results: Updates

Understanding of incentives did not affect direction of updates

figures/update/update_dir_correct_by_prior_udst.pdf